

MAT 314 HW03

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Problem (1.2). Prove that, for $n \neq 0$, $\cos(2\pi/n)$ is an algebraic number.

Solution. We know that

$$(\cos(2\pi/n) + i \sin(2\pi/n))^n = \cos(2\pi) + i \sin(2\pi) = 1.$$

Let $x = \cos(2\pi/n)$. Then, the left-hand side expands to

$$\sum_{k=0}^n \binom{n}{k} x^{n-k} (\sqrt{1-x^2})^k \cdot i^k.$$

The real part of this quantity is the sum of the addends for which i^k is real, i.e. k is even, which amounts to

$$\sum_{0 \leq k' \leq n/2} \binom{n}{2k'} x^{n-2k'} (1-x^2)^{k'} \cdot (-1)^{k'} = 1.$$

Thus $\cos(2\pi/n)$ is a root of an integer polynomial

$$P(x) = \sum_{0 \leq k' \leq n/2} \binom{n}{2k'} x^{n-2k'} (1-x^2)^{k'} \cdot (-1)^{k'} - 1,$$

which is what it means for $\cos(2\pi/n)$ to be algebraic. $\square$

Problem (2.1). For which positive integers n does $x^2 + x + 1$ divide $x^4 + 3x^3 + x^2 + 7x + 5$ in $[\mathbb{Z}/(n)][x]$?

Solution. Let $P(x) = x^4 + 3x^3 + x^2 + 7x + 5$. The division algorithm gives

$$P(x) = (x^2 + x + 1)(x^2 + 2x - 2) + 7x + 7.$$

We want the remainder to be zero in $[\mathbb{Z}/(n)][x]$, which means that $7x + 7 = nQ(x)$ for some $Q(x) \in [\mathbb{Z}/(n)][x]$. The only choice for n here is thus $\boxed{7}$. $\square$

Problem (3.5). The derivative of a polynomial f with coefficients in a field F is defined by the formula

$$(a_n x^n + \cdots + a_1 x + a_0)' = n a_n x^{n-1} + \cdots + 1 a_1.$$

- (a) Prove the product rule $(fg)' = f'g + fg'$ and the chain rule $(f \circ g)' = (f' \circ g) \cdot g'$.
- (b) Let α be an element of F . Prove that α is a multiple root of a polynomial f if and only if it is a common root of f and f' .

Solution. (a) Let $f(x) = a_n x^n + \cdots + a_1 x + a_0$ and $g(x) = b_m x^m + \cdots + b_1 x + b_0$. Then we observe that for any monomial cx^k ,

$$\begin{aligned} (f(x) \cdot cx^k)' &= (ca_n x^{k+n} + \cdots + ca_1 x^{k+1} + ca_0 x^k)' \\ &= \sum_{j=0}^n c(k+j) a_j x^{k+j-1} \\ &= cx^k \sum_{j=0}^n (j a_j x^{j-1}) + ckx^{k-1} \sum_{j=0}^n a_j x^j \\ &= (cx^k) f'(x) + (cx^k)' f(x), \end{aligned}$$

so the product rule holds when one of the polynomials is a monomial. We can clearly see that the derivative is a linear operator on $F[x]$, so this gives us

$$\begin{aligned}(fg)' &= \sum_{j=0}^m (f(x) \cdot b_j x^j)' = \sum_{j=0}^m f'(x) b_j x^j + f(x) \cdot j b_j x^{j-1} \\ &= f'(x) \sum_{j=0}^m b_j x^j + f(x) \sum_{j=0}^m j b_j x^{j-1} \\ &= f'g + fg'.\end{aligned}$$

We now prove the chain rule. First, we will prove the chain rule when $f(x) = cx^k$ is a monomial by induction on k . If $k = 0$ then f is constant, so $f \circ g$ is constant and hence its derivative is zero, which aligns with $f' \circ g = 0$. Now suppose that the chain rule holds for monomials with degree $k - 1$. Then if $f(x) = cx^k$,

$$\begin{aligned}(f \circ g)' &= (c \cdot g^k)' = c(g^{k-1} \cdot g)' \\ &= c \cdot ((g^{k-1})' \cdot g + g^{k-1} \cdot g') \\ &= c \cdot ((k-1)g^{k-2} \cdot g' \cdot g + g^{k-1} \cdot g') \\ &= c \cdot k g^{k-1} \cdot g',\end{aligned}$$

where the second-to-last equality comes from applying the chain rule to the composition $x \mapsto g(x) \mapsto g(x)^{k-1}$ via the induction hypothesis. To finish, we can extend to arbitrary f via linearity:

$$\begin{aligned}(f \circ g)' &= \left(\sum_{j=0}^n a_j g^j \right)' = \sum_{j=0}^n (a_j g^j)' \\ &= \sum_{j=0}^n j a_j g^{j-1} \cdot g' = g' \cdot (f' \circ g).\end{aligned}$$

(b) Suppose α is a multiple root of f . Then $(x - \alpha)^2 \mid f$, so, writing $f(x) = (x - \alpha)^2 g(x)$,

$$f'(x) = (x - \alpha)^2 g'(x) + 2(x - \alpha)g(x),$$

which is clearly divisible by $x - \alpha$, hence f' has α as a root.

Conversely, suppose α is a common root of f and f' . Writing $f(x) = (x - \alpha)g(x)$, we get

$$f'(x) = (x - \alpha)g'(x) + g(x),$$

so setting $x = \alpha$ yields

$$0 = f'(\alpha) = g(\alpha),$$

hence $x - \alpha \mid g(x)$ implying $(x - \alpha)^2 \mid f(x)$ as desired. $\square$

Problem (5.7). Let F be a field and let $R = F[t]$ be the polynomial ring. Let R' be the ring extension $R[x]/(tx - 1)$ obtained by adjoining an inverse of t to R . Prove that this ring can be identified as the ring of *Laurent polynomials*, which are finite linear combinations of powers of t , negative exponents included.

Solution. Let $F[t, t^{-1}]$ be the ring of Laurent polynomials in t over F . We wish to show that $F[t, x]/(tx - 1) \cong F[t, t^{-1}]$, which we will do by showing that the map $\phi: F[t, x] \rightarrow F[t, t^{-1}]$ given by

$$\phi \left(\sum_{k=0}^n \sum_{i=0}^k a_{i,k-i} t^i x^{k-i} \right) = \sum_{k=0}^n \sum_{i=0}^k a_{i,k-i} t^{2i-k}$$

is a homomorphism of rings with kernel $(tx - 1)$. We can see easily that $\phi(1) = 1$ and $\phi(p + q) = \phi(p) + \phi(q)$ for any $p, q \in F[t, x]$. To show that $\phi(pq) = \phi(p)\phi(q)$, we only need to prove this in the

case that one of p, q is a monomial (WLOG q), since we can get at general polynomials by linearity. The case where q is a monomial is easy: if $q(t, x) = ct^r x^s$, then

$$\phi(pq) = \phi\left(\sum_{k=0}^n \sum_{i=0}^k ca_{i,k-i} t^{i+r} x^{k-i+s}\right) = \sum_{k=0}^n \sum_{i=0}^k a_{i,k-i} t^{2i-k} \cdot ct^{r-s} = \phi(p) \cdot \phi(q)$$

since $\phi(q) = ct^{r-s}$. To be more precise about general q , we note that $\phi(pq)$ can be expanded as a sum of terms of the form $\phi(pq_{i,j})$ where $q_{i,j}$ is the monomial $b_{i,j} t^i x^j$ in q , and then we can write this as $\phi(p)\phi(q_{i,j})$ and use the distributive property and the additivity of ϕ to recombine the $\phi(q_{i,j})$ into $\phi(q)$.

Now we show that $\ker \phi = (tx - 1)$. If

$$\phi\left(\sum_{k=0}^n \sum_{i=0}^k a_{i,k-i} t^i x^{k-i}\right) = \sum_{k=0}^n \sum_{i=0}^k a_{i,k-i} t^{2i-k} = 0,$$

we can rewrite

$$\sum_{k=0}^n \sum_{i=0}^k a_{i,k-i} t^i x^{k-i} = \sum_{d=1}^n \sum_{i=0}^{n-d} a_{d+i,i} t^{d+i} x^i + \sum_{d=1}^n \sum_{i=0}^{n-d} a_{i,d+i} t^i x^{d+i} + \sum_{i=0}^n a_{i,i} t^i x^i.$$

Then the image of this polynomial under ϕ being zero is equivalent to

$$\begin{aligned} \phi\left(\sum_{i=0}^{n-d} a_{d+i,i} t^{d+i} x^i\right) &= t^d \sum_{i=0}^{n-d} a_{d+i,i} = 0 \\ \phi\left(\sum_{i=0}^{n-d} a_{i,d+i} t^i x^{d+i}\right) &= x^d \sum_{i=0}^{n-d} a_{i,d+i} = 0 \\ \phi\left(\sum_{i=0}^n a_{i,i} t^i x^i\right) &= \sum_{i=0}^n a_{i,i} = 0. \end{aligned}$$

I claim that this implies each of the arguments of ϕ above are divisible by $tx - 1$, which is enough to imply that the kernel is $(tx - 1)$ (since clearly any polynomial with $tx - 1$ as a factor gets sent to zero because $\phi(tx - 1) = t \cdot t^{-1} - 1 = 0$). Indeed

$$\sum_{i=0}^{n-d} a_{d+i,i} t^{d+i} x^i = t^d \cdot \sum_{i=0}^{n-d} a_{d+i,i} t^i x^i$$

is t^d times a polynomial in tx whose coefficients sum to zero; the theory of single-variable polynomials thus tells us that $tx = 1$ being a root implies that $tx - 1$ is a factor of this polynomial. Similar logic applies to the other two equations (the second after factoring x^d instead), so our original polynomial that got sent to zero can be written as a sum of polynomials divisible by $tx - 1$, which implies the result. $\square$

Problem (6.8). Let I and J be ideals of a ring R such that $I + J = R$.

- Prove that $IJ = I \cap J$.
- Prove the *Chinese Remainder Theorem*: For any pair a, b of elements of R , there is an element x such that $x \equiv a \pmod{I}$ and $x \equiv b \pmod{J}$.
- Prove that if $IJ = 0$, then R is isomorphic to the product ring $(R/I) \times (R/J)$.
- Describe the idempotents corresponding to the product decomposition in (c).

Solution. (a) On one hand $IJ \subseteq I \cap J$ because if $i \in I$ and $j \in J$ then $ij \in I$ and $ij \in J$ both hold by definition of ideal. On the other hand, suppose that $a \in I \cap J$. Then we can write $1 = i + j$ for some $i \in I$, $j \in J$, so that $ai + aj \in IJ$ because ai and aj are both products of an element of I with an element of J .

(b) As before write $1 = i + j$ and set $x = aj + bi$. Then

$$x \equiv aj = a(1 - i) \equiv a \pmod{I}, \quad x \equiv bi = b(1 - j) \equiv b \pmod{J}$$

as desired.

(c) Let $\phi_I: R \rightarrow R/I$ and $\phi_J: R \rightarrow R/J$ be the projection homomorphisms. Consider the homomorphism $\phi: R \rightarrow (R/I) \times (R/J)$ given by

$$\phi(x) := (\phi_I(x), \phi_J(x)).$$

We have $x \in \ker \phi$ if and only if $\phi_I(x) = \phi_J(x) = 0$, so the kernel is $I \cap J = IJ = 0$.

(d) Let $e \in R$ be the element such that $e \equiv 1 \pmod{I}$ and $e \equiv 0 \pmod{J}$. We can see that e is idempotent. Moreover, $ex \equiv 0 \pmod{I}$ if and only if $x \in I$; the only if direction is because if $ex \in I$ and $x = i + j$ for $i \in I$, $j \in J$, then $ex \equiv ej \equiv j \pmod{I}$, implying $j \in I \cap J = IJ = 0$ so $x \in I$. Thus the kernel of the map $f: R \rightarrow R/I$ given by $f(x) = ex$ is I , so $R/I \cong eR$ and similarly $R/J \cong (1 - e)R$, giving us

$$R \cong eR \times (1 - e)R. \quad \square$$